Olist Delivery Reliability and Customer Reviews

SQL and Power BI case study

Brazilian E-Commerce Public Dataset by Olist, 2016–2018

Main question

Which delivery routes produced more late orders than expected, and how were late deliveries related to customer reviews?

Summary

Late orders received much worse reviews than on-time orders. A small number of interstate routes produced most late deliveries. After month and distance were taken into account, SP → RJ remained the largest problem route. On the priority routes, most of the additional time appeared after carrier handoff.

Delivery Reliability Executive Overview

96470	6534	62,42	2,27
Analyzed orders	Late orders	% Bad review rate among late orders	Average review score among late orders

Route	Adjusted excess late orders	Actual late rate, %	Expected late rate, %
SP → RJ	689,40	14,25	5,65
SP → BA	104,40	12,89	8,28
SP → ES	75,40	12,33	7,07
SP → SC	48,70	9,57	7,45
PR → RJ	39,00	12,34	8,22

1. Data and method

The source contains orders, customers, order items, sellers, payments, reviews, products and geolocation data. The analysis was built at order level.

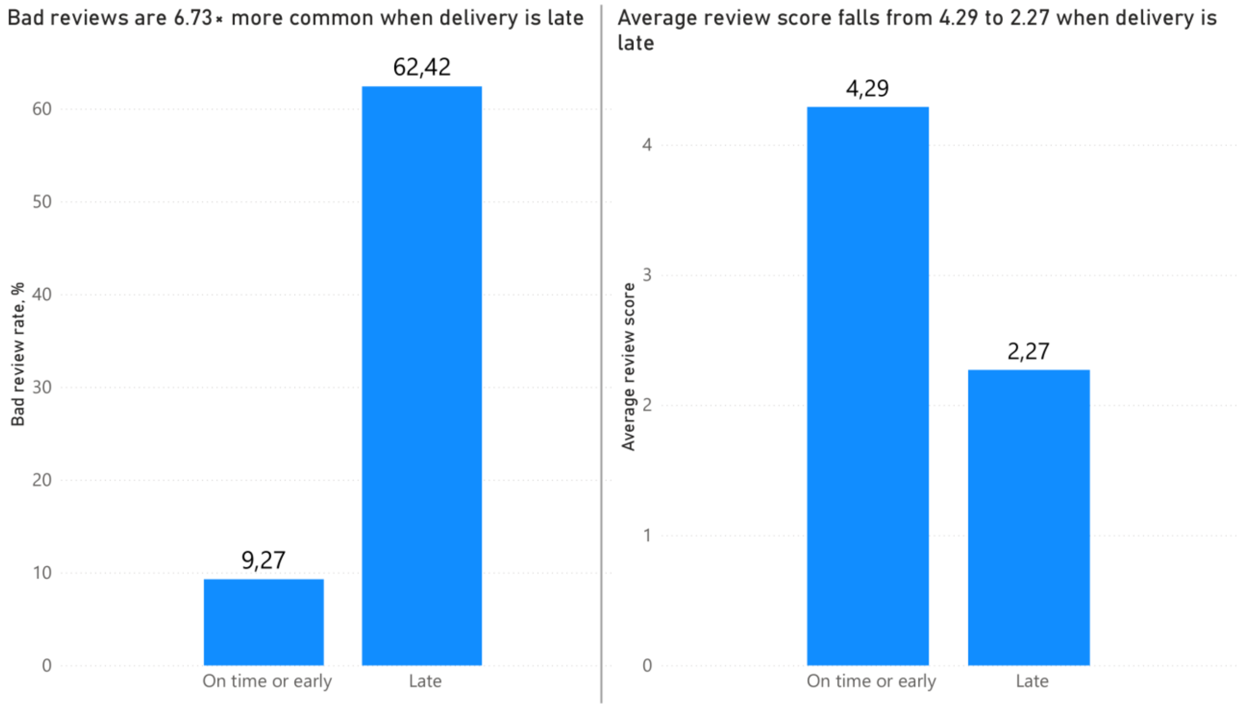
Measure	Value
Delivered orders in the main analysis	96,470
Orders with reviews	95,824
Late orders (1+ day)	6,534
Late delivery rate (1+ day)	6.77%
Bad review rate	12.81%
Single-seller interstate orders	60,961
Interstate orders with distance	60,551 (99.33%)

- For the headline comparison, a delivery is late when it arrives at least one full day after the estimate; delays of less than one day are shown separately in Section 3.
- A bad review means a score of 1 or 2.
- Order items and payments were aggregated before joining to orders.
- The latest review row was kept for each order.
- Route analysis uses single-seller orders so that each order has one origin state.
- Distance is the straight-line distance between average ZIP-prefix coordinates.

2. Late delivery and customer reviews

Delivery group	Orders	Reviewed orders	Average score	Bad review rate
On time, early, or <1 day late	89,936	89,443	4.29	9.27%
At least 1 day late	6,534	6,381	2.27	62.42%

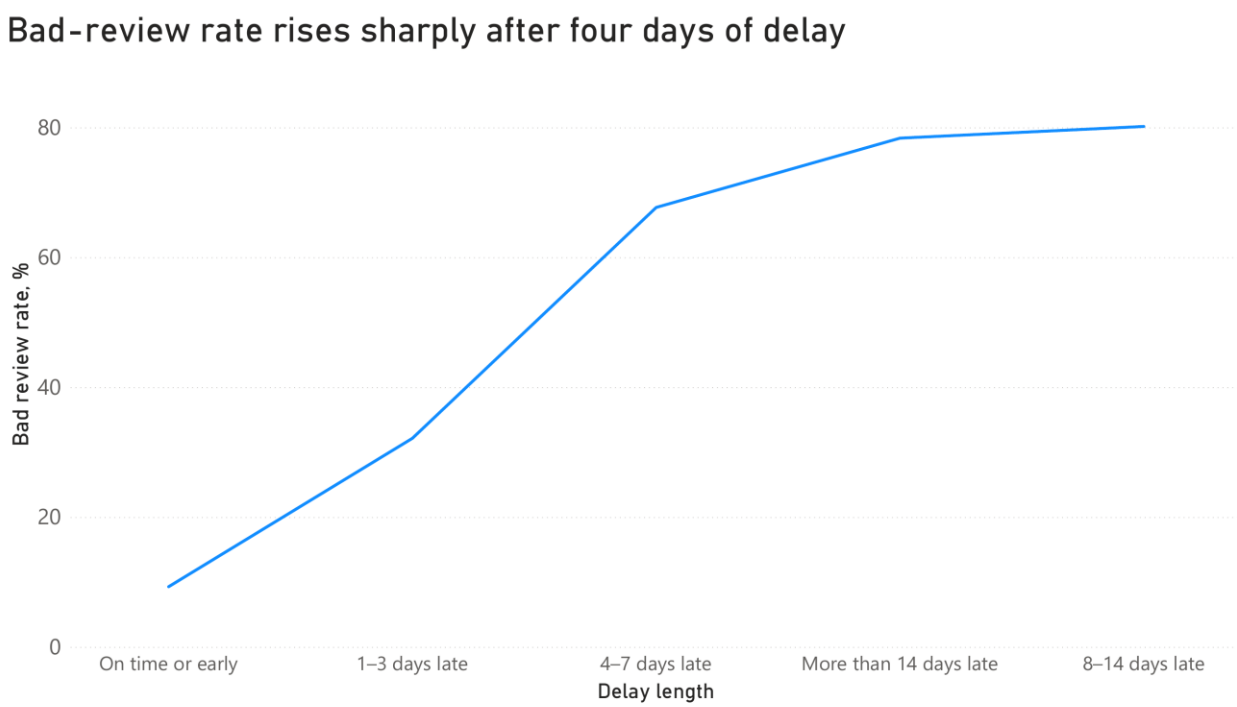
A bad review was 6.73 times more common among orders delayed by at least one day. The average score fell from 4.29 to 2.27.



3. Delay length

Delay group	Reviewed orders	Average score	Bad review rate
On time or early	88,163	4.29	9.22%
Less than 1 day late	1,280	4.03	12.42%
1–3 days late	1,852	3.29	32.13%
4–7 days late	1,748	2.10	67.68%
8–14 days late	1,446	1.67	80.15%
More than 14 days late	1,335	1.72	78.35%

The result changed sharply after about four days. For orders delayed by 4–7 days, 67.68% of reviews were bad.



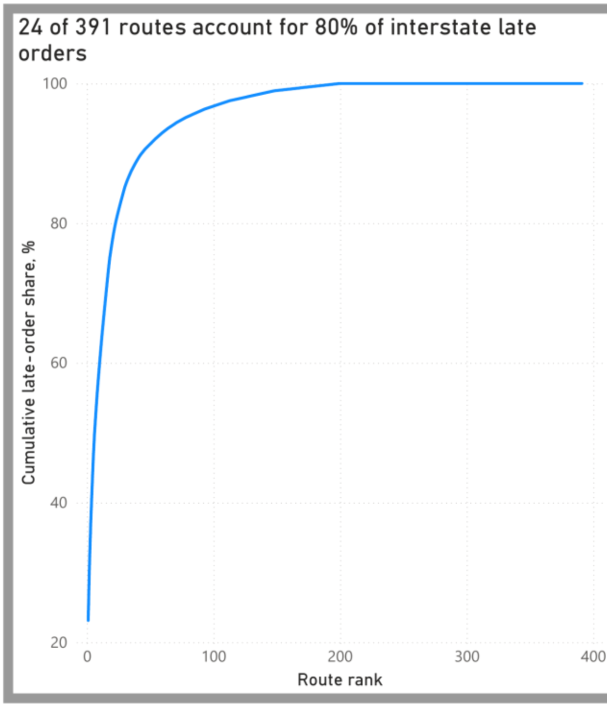
4. Route concentration

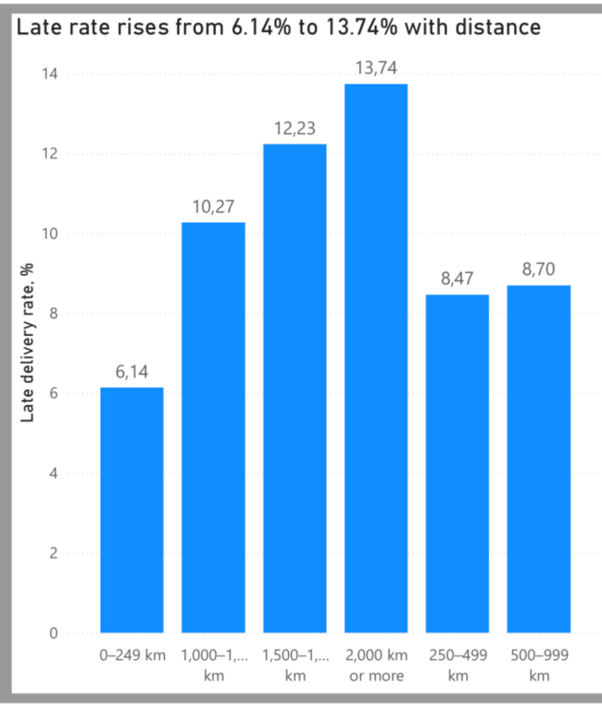
There were 391 interstate route pairs. Twenty-four routes, or 6.14% of all route pairs, produced at least 80% of interstate late orders. SP → RJ alone produced about 22% of interstate late orders.

5. Distance and late delivery

Distance	Orders	Late rate	Average delivery	Median carrier time
0–249 km	1,970	6.14%	11.68 days	172 hours
250–499 km	18,861	8.47%	12.75 days	175 hours
500–999 km	24,528	8.70%	14.04 days	214 hours
1,000–1,499 km	6,487	10.27%	16.88 days	285 hours
1,500–1,999 km	3,132	12.23%	19.01 days	330 hours
2,000 km or more	5,573	13.74%	20.75 days	355 hours

Longer deliveries were more likely to be late. Route performance was therefore compared with other routes in the same month and distance band.


24 of 391 routes account for 80% of interstate late orders
Cumulative late-order share, %
100
80
60
40
20
0
100
200
300
400
Route rank

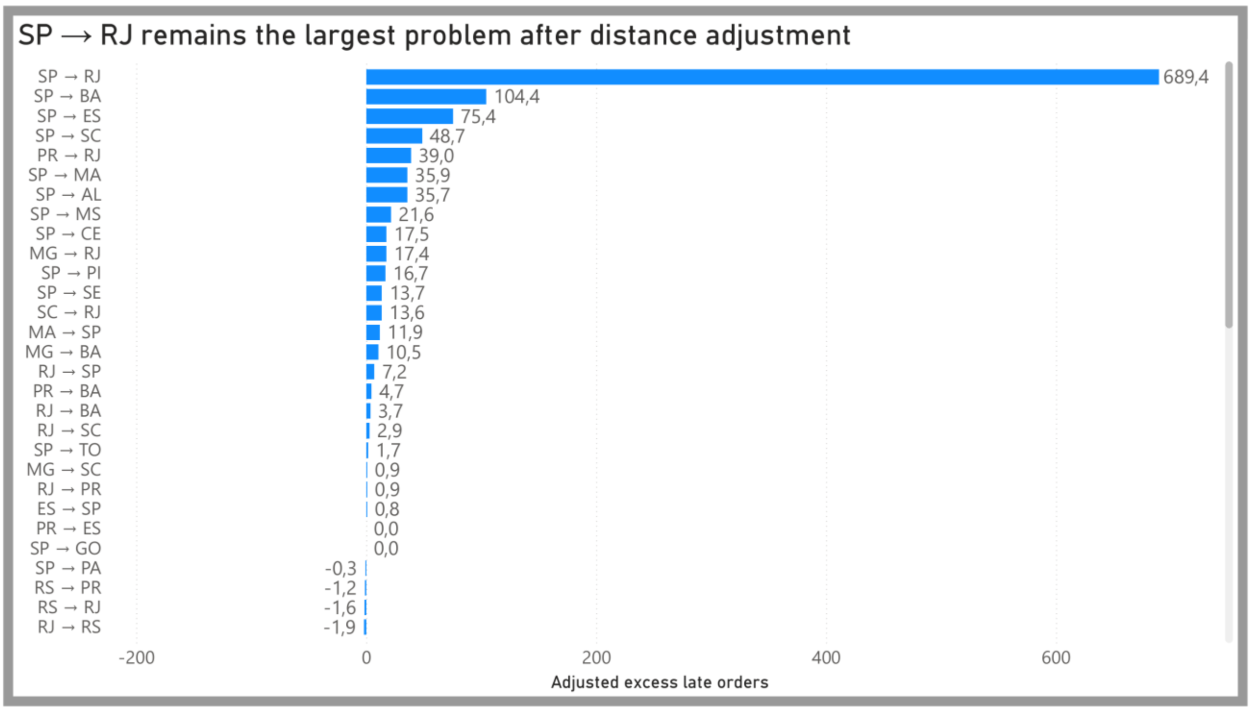

Late rate rises from 6.14% to 13.74% with distance
Late delivery rate, %
14
12
10
8
6
4
2
0
6,14
10,27
12,23
13,74
8,47
8,70
0–249 km
1,000–1,... km
1,500–1,... km
2,000 km or more
250–499 km
500–999 km

6. Routes with more late orders than expected

Expected late orders were calculated from other routes in the same month and distance band. The route itself was excluded from its benchmark.

Route	Orders	Actual late	Expected late	Excess late	Actual vs expected
SP → RJ	8,021	1,143	453.6	689.4	14.25% vs 5.65%
SP → BA	2,265	292	187.6	104.4	12.89% vs 8.28%
SP → ES	1,436	177	101.6	75.4	12.33% vs 7.07%
SP → SC	2,299	220	171.3	48.7	9.57% vs 7.45%
PR → RJ	948	117	78.0	39.0	12.34% vs 8.22%

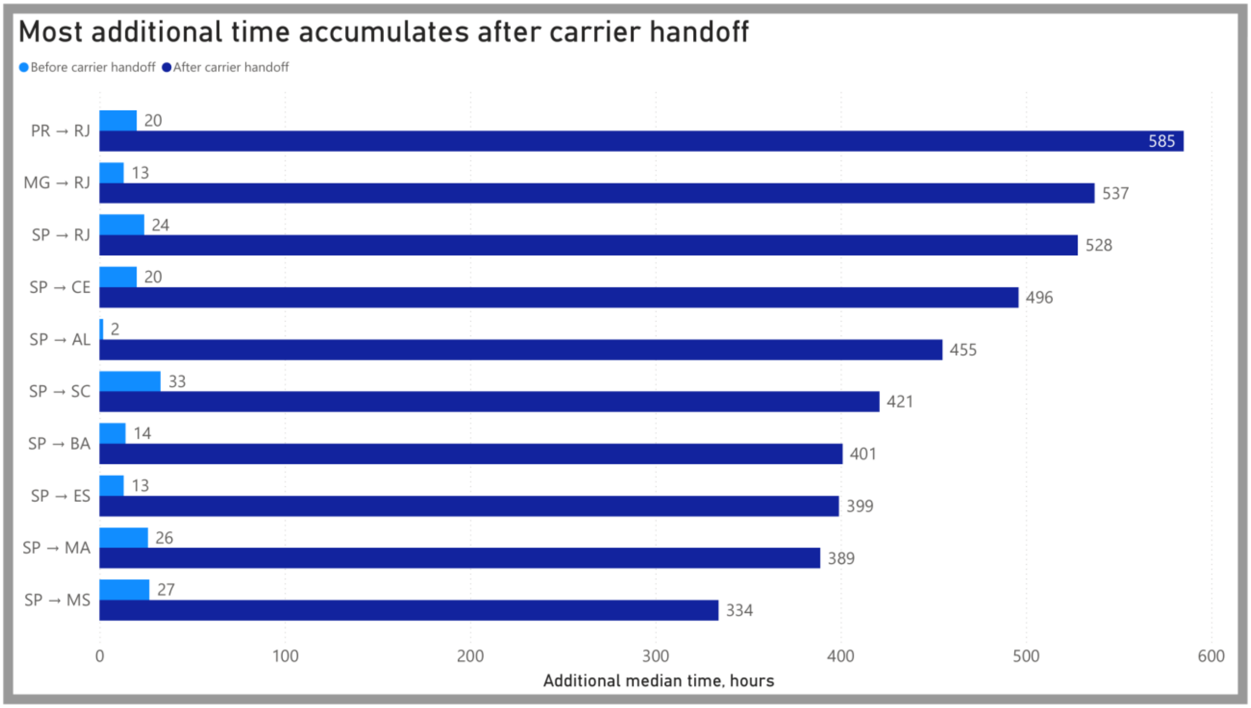
SP → RJ remained the largest problem after adjustment. SP → PA no longer looked poor after distance was taken into account.



7. Where the delay accumulates

Group	Median time before carrier	Median time after carrier handoff
On-time orders	42 hours	165 hours
Late orders	72 hours	575 hours
Difference	+30 hours	+410 hours

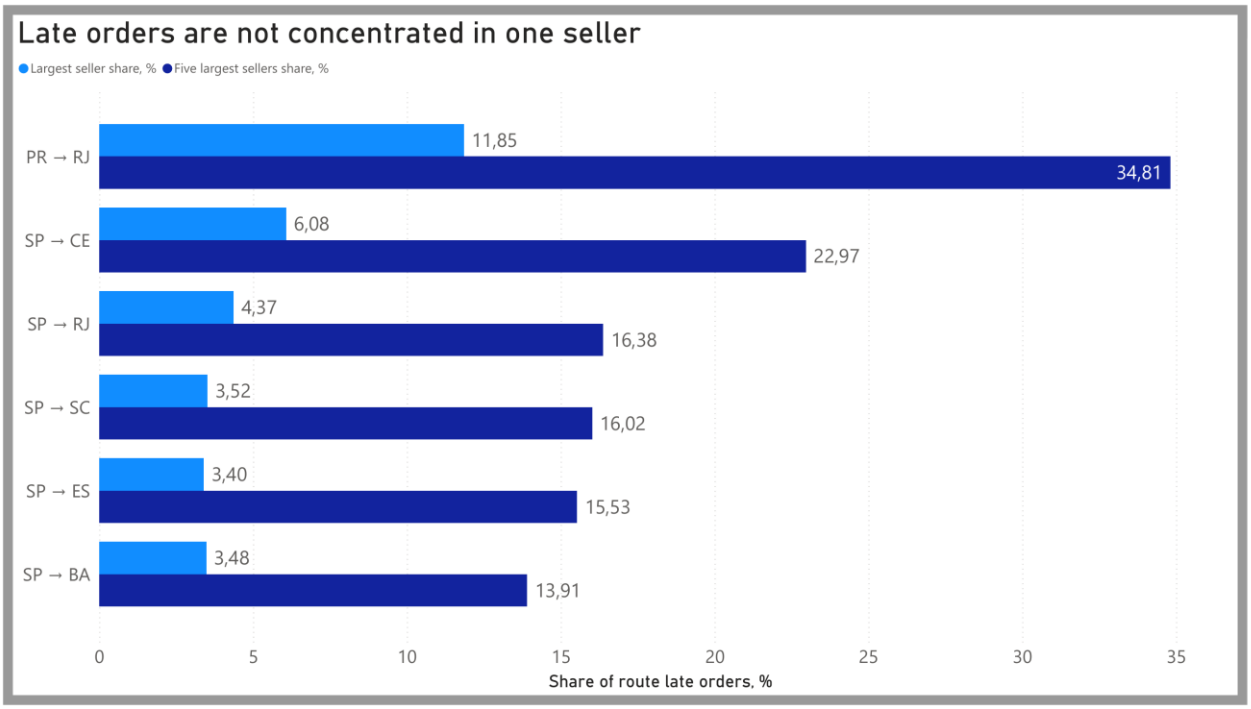
On the ten priority routes, the additional median time before carrier handoff ranged from 2 to 33 hours. The additional time after carrier handoff ranged from 334 to 585 hours.



8. Seller concentration

The problem was not concentrated in one seller. Across the ten priority routes, the largest seller produced 3.40%–11.85% of late orders. The five largest sellers produced 13.91%–34.81%.

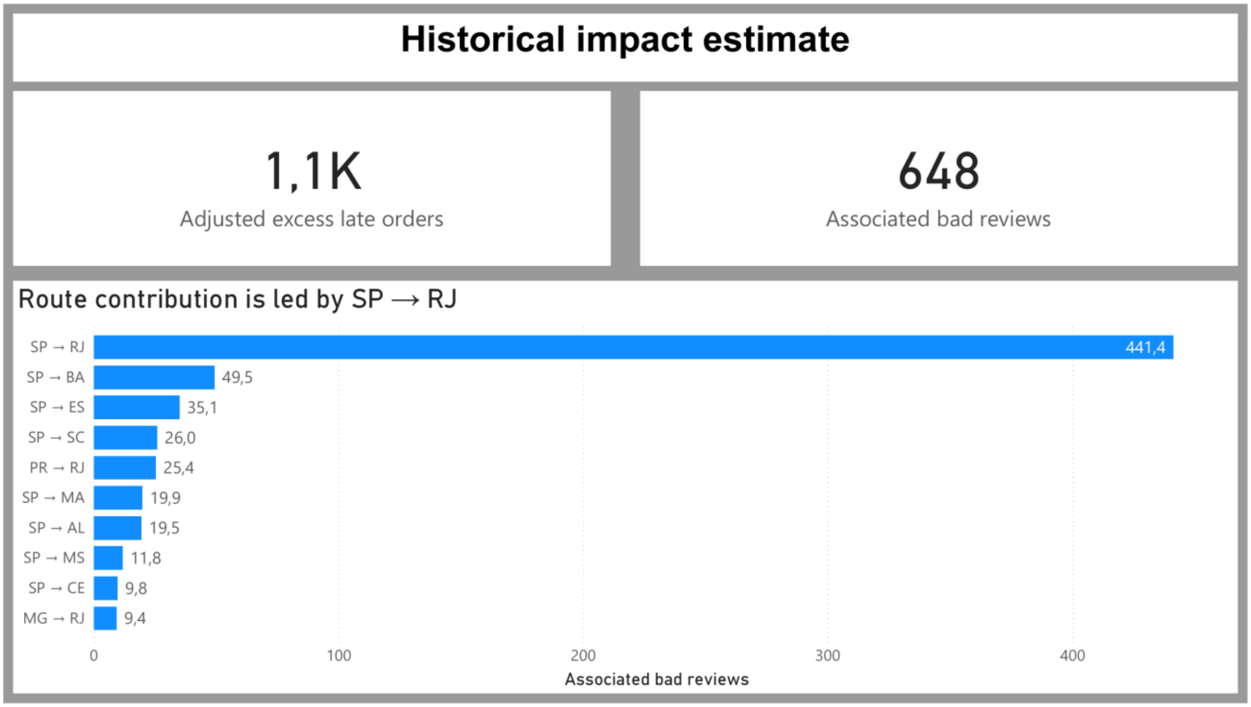
Route	Seller count	Top seller share	Top five sellers share
SP → RJ	1,002	4.37%	16.38%
SP → BA	594	3.48%	13.91%
SP → ES	480	3.40%	15.53%
SP → CE	344	6.08%	22.97%
PR → RJ	178	11.85%	34.81%
SP → SC	589	3.52%	16.02%



9. Historical impact estimate

The ten largest adjusted problem routes produced 1,085.0 late orders above their month-and-distance benchmarks. Using the observed difference in bad-review rates between late and comparison-group orders within each route gives an estimate of about 648 associated bad reviews.

This is a historical comparison, not a guaranteed forecast and not a causal estimate.



10. Recommendations

1. Investigate SP → RJ first. It has 689.4 late orders above its benchmark and an average distance of 441 km.
2. Check carrier handoff, intermediate hubs and service-level performance on SP → RJ, SP → BA and SP → ES.
3. Track actual late rate, expected late rate, excess late orders, order volume and bad review rate by route each month.
4. Contact customers proactively when an order approaches four days late.
5. Use excess late-order count to set priorities. Use excess percentage points to find smaller high-risk routes.
6. Test changes on a small number of routes and compare them with similar routes before making causal claims.